Preliminary.

The Zorn's Lemma gives several facts

1). Given a field F, there exists the Algebraic Closure $\overline{F}$.

2). Let F be a field and E/F is an Algebraic extension.
Given an Embbeding $\varphi: F \to L$ where L is algebraically closed field, there exists an Extension $\tilde{\varphi}: E \to L$ of φ.
Moreover, if E is algebraically closed, such $\tilde{\varphi}$ is Isomorphism.

In general, field Homomorphisms don't need to behave 'good' properties.
For instance, Let $K = \mathbb{Q}(\sqrt[3]{2})$. The minimal polynomial x^3-2 over $\mathbb{Q}$ of $\sqrt[3]{2}$ has three distinct roots, $\sqrt[3]{2}$, $w \cdot \sqrt[3]{2}$, $w^2 \cdot \sqrt[3]{2}$ where w is primitive 3th root of unity. Therefore, $\# \mathrm{Emb}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}), \mathbb{C}) = [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$.
But, since $w^i \cdot \sqrt[3]{2} \notin \mathbb{Q}(\sqrt[3]{2})$ $i=1,2$, so

$$\mathrm{Aut}(\mathbb{Q}(\sqrt[3]{2})) = \{id\}.$$